Week 10 23.03.2026

Strategic manipulation in elections

The voting model includes:

- Set of voters $N = \{1, \dots, n\}$ $(n \geq 2)$
- Set of candidates $C = \{c_1, \dots, c_m\}$ $(m \geq 2)$
- Each voter $i \in N$ has a ranking $R \in \mathcal{L}(C)$ over the alternatives, where $x \succ_{R_i} y$ means that voter i prefers x to y
- A preference profile $R \in \mathcal{L}(C)^n$ is a collection of all voters' rankings
- A social choice function
$$F : \mathcal{L}(C)^n \to C$$

[today, we only deal with resolute voting rules; these can be obtained from irresolute rule with the help of tie-breaking schemes]

So far we have assumed voters are honest and represent their preferences as they truly are. However, voters can manipulate their votes strategically to achieve better outcomes. To accommodate this, we now have to distinguish:

- the ballot a voter reports
- her actual preference order

Both are elements of $\mathcal{L}(C)$. If they coincide, then the voter is <u>truthful</u>.

F is <u>strategyproof</u> (or <u>immune to manipulation</u>) if for no voter $i \in N$ there exist a profile R (including i's truthful preference R_i) and an untruthful ballot R_i' for i such that R_i ranks $F(R_i', R_{-i})$ above $F(R)$.

Thus: nobody has an incentive to misrepresent their preferences.

<u>Example 1 (Manipulation in the Borda count)</u>

Consider the following true preference profile:

1	2	3
b	b	a
a	a	b
c	c	c
d	d	d

Here, b has 8 points, a has 7 points, and c and d are both weaker, and so b wins under the Borda count.
Now, what if voter 3 changes their vote, resulting in the following preference profile:

1	2	3
b	b	a
a	a	c
c	c	d
d	d	b

Now, a still has 7 points, but b has 6 points. Since c and d have even fewer points, a would be the winner of this election. Thus, voter 3 manipulated their vote and misrepresented their true ranking to result in a better outcome for themselves, because, in reality, they wanted a to win.

<u>Example 2 (When is plurality strategyproof?)</u>

It turns out that plurality is only strategyproof for a maximum of 2 alternatives.

For 2 alternatives, the only deviation a voter can make is flip their second favourite alternative with their top choice, which directly contributes to the second-favourite alternative's count, resulting in the same or a worse outcome for the voter.

For 3 alternatives, however, if a voter's top choice has no chance of winning but their second and third choices are tied in plurality scores, that voter is better off misreporting their true preference by reporting their second choice to ensure their second choice wins over their third choice.

1	2	3	4	5
a	a	b	b	c
b	b	a	a	b
c	c	c	c	a

$\rightarrow$

1	2	3	4	5
a	a	b	b	b
b	b	a	a	c
c	c	c	c	a

Gibbard-Satterthwaite theorem

A resolute SCF F is <u>onto</u> (or <u>surjective</u>) if for any alternative, a ~~strategy~~ preference profile exists that will result in the alternative getting selected. That is, for any $x \in C$ there is a profile R such that $F(R) = x$.

A voting rule F is <u>dictatorial</u> if there is a voter that always get their top choice, no matter what other voters' want.

<u>Example: dictatorship</u>

- 2 voters: the row player is a dictator
- 3 candidates: $\{a, b, c\}$

	a	b	c
a	a	a	a
b	b	b	b
c	c	c	c

G-S theorem: Let $m \geq 3$.

A social choice function F is strategyproof (SP) and onto C if and only if F is a dictatorship.

This theorem implies that any voting rule that is onto and nondictatorial is manipulable.

For $m \geq 3$, all common rules are onto and nondictatorial. What if we enforce SP and nondictatorial but relax the onto constraint?

- The rule that says "regardless of anyone's votes, pick alternative a" is SP and non-dictatorial, because no one's vote matters. Further, the rule that says "in advance, we decide that either a or b wins, and we'll take the

majority between them" is also SP and nondictatorial.

<u>Proof sketch of G-S theorem:</u>

If F is dictatorial, then trivially F is onto because the dictator can rank any alternative first, and F is also SP because the dictator is best-off telling their true preference because it will be the result of the election, and the reported preferences of all other voters do not matter.
Proving the other direction is far more interesting.
We will start with the following lemmas:

- <u>Strong monotonicity</u>: If F is a SP function and R is some profile where $F(R) = a$, then $F(R') = a$ for all profiles

R' such that $\forall x \in C, i \in N$:

$$a \succ_{R_i} x \Rightarrow a \succ_{R'_i} x.$$

- <u>Unanimity</u>: If F is a SP and onto function, and R is a profile where $a \succ_{R_i} b$ for all $i \in N$, then $F(R) \neq b$.

Strong monotonicity can be interpreted as the following: if F is strategyproof then if the outcome of some profile is a, then if you make a new profile where everything that was ranked below a in voter preferences stays ranked below a (regardless of where the alternatives above a are ranked), then the outcome of the election will still be a.

Unanimity can be interpreted as the following: If F is SP and onto, then if everyone

ranks a above b then b will not win the election.

Let us also assume that $m \geq n$ and neutrality, that states that for any permutation of alternatives $\pi : C \to C$,

$$F(\pi(R)) = \pi(F(R)).$$

That is to say, if a was the winner of an election and we permute the alternatives in the profile such that some alternative b takes a's place in every voter's ranking, then the winner from this new profile will be b.

Note that neutrality and $m \geq n$ are assumptions that are not necessary for the G-S theorem, but they make the proof more simple.

We will even further simplify the proof for $m=5$ and $n=4$. It is generalisable for $m \geq n$, from here.

Assume that F is SP and onto.
Consider the profile:

$$R = \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \hline a & b & c & d \\ b & c & d & a \\ c & d & a & b \\ d & a & b & c \\ e & e & e & e \end{array}$$

Note here all voters rank e last, and alternatives a trough d are ranked in a cyclic fashion across the voters. By unanimity, it is clear that e cannot be the winner. Since alternatives a through d are all symmetric across the preferences, WLOG assume $F(R) = a$.
Now, consider the following profile, created matching the rankings of voters 2 and 3 to the ranking of voter 4 and by promoting candidate

d to the 2nd position in the ranking of voter 1:

$$R^1 = \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \hline a & d & d & d \\ d & a & a & a \\ b & b & b & b \\ c & c & c & c \\ e & e & e & e \end{array}$$

By strong monotonicity, it follows that $F(R^1) = a$. Strong monotonicity applies because in all the preferences that have changed, the alternatives ranked below a in R are still below a in R^1. Now, consider the following profile constructed by changing voter 2's ranking by dropping a to their least favourite preference:

$$R^2 = \begin{array}{cccc} 1 & 2 & 3 & 4 \\ \hline a & d & d & d \\ d & b & a & a \\ b & c & b & b \\ c & e & c & c \\ e & a & e & e \end{array}$$

Note that in this profile, alternative d dominates alternatives b, c, e, so the winner cannot be either of these three alternatives. Further, since F is SP, the winner cannot be d because otherwise voter 2 would have had a useful misrepresentation if R_2^1 was their true preference, as they would prefer d over a and by misrepresenting to R_2^2 they could change the result from a to d. Thus, from the process of elimination, we conclude that $F(R^2) = a$.

Similarly, if we construct R^3 by changing voter 3's ranking by dropping a to their least favourite preference, we will get $F(R^3) = a$ by the same rationale. If we construct R^4 by further editing voter 4's preference, we will get $F(R^4) = a$ where

1	2	3	4
a	d	d	d
d	b	b	b
b	c	c	c
c	e	e	e
e	a	a	a

From here, we claim that strong monotonicity implies that $F(R') = a$ for every R' where voter 1 ranks a first. That is because if voter 1 ranks a first in R', then all alternatives

that were ranked below a in R^4 are ranked below a in R' (because no alternatives are ranked below a in the preferences of voters 2, 3, 4), and so strong monotonicity says that $f(R') = a$ as well. We now claim that neutrality implies that voter 1 is a dictator. This is because for any profile R', if voter 1 ranks alternative x first in R', then we can construct a permutation π that swaps x with a. Since we have just shown that $f(\pi(R')) = a$ (voter 1 ranks a first in $\pi(R')$), it follows from neutrality that $\pi(F(R')) = a$ and so $F(R') = x$, implying that voter 1 is a dictator because whatever they ranked first won the election! Note that it originally $F(R)$ was b, c or d,

we would have shown that voter 2, 3 or 4 was the dictator, respectively. □

Barriers to strategic manipulation

Manipulations may be unavoidable, but is that the end of it?

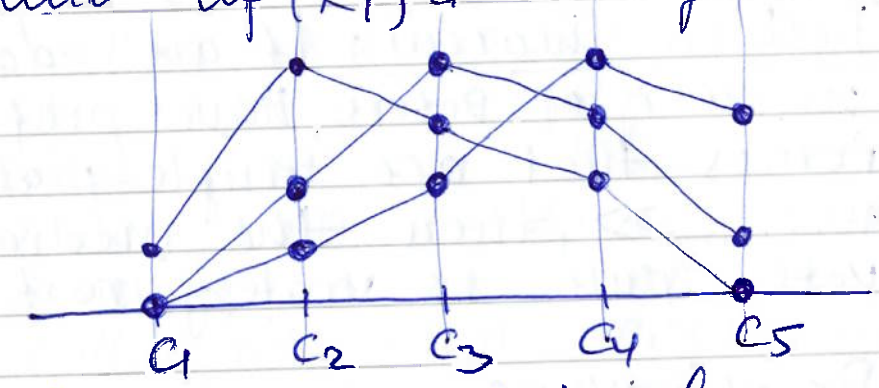
No!

We shall now briefly review two kinds of barriers against strategic manipulation.

① Domain restriction: single-peaked preferences

Every voting rule can be manipulated, but not in all profiles. Can we do better if we restrict attention to specific (natural) profiles? We only discuss the oldest and most famous domain restriction.

A profile $R = (R_1, \ldots, R_n)$ is <u>single-peaked</u> if we can arrange the alternatives from left to right along some dimension $\gg$ such that R_i ranks x above y whenever x is between y and $top(R_i)$ according to $\gg$.



Sometimes a natural assumption: traditional political parties, spatial voting, agreeing on a number (e.g., legal drinking age), etc.

<u>Strategyproofness of the Median-Voter Rule</u>

For a given dimension $\gg$, the median-voter rule asks each voter for her top alternative and elects the alternative proposed by the voter corresponding to the median w.r.t. $\gg$.

<u>Folklore theorem</u>: If an odd number of voters have preferences that are single-peaked w.r.t. $\gg$, then the median-voter rule is strategyproof.

<u>Proof</u>: WLOG, our manipulator's top alternative is to the right of the median (the winner). If she declares a peak further to the right, nothing will change. If she declares a peak further to the left, either nothing will change, or the new winner will be even worse! $\square$

This is closely related to Black's Median voter theorem, showing that under the same conditions a Condorcet winner exists and is elected.

Intuitively, since in single-peaked preferences alternatives lie on a line where each voter has one most preferred point (their peak) and their preference strictly falls as you move away from that peak in either direction, majority cycles cannot occur, so a Condorcet winner always exists (for an odd number of voters) and is the candidate at the median peak.

② Computational barriers to manipulation

Every voting rule can be mani-

pulated in some profiles. But even when it is possible to manipulate, maybe actually doing so is difficult?

Tools from complexity theory can help make this idea precise.

If manipulation is computationally intractable for F, then we might consider F <u>resistant</u> (but not immune) to manipulation.

The <u>computational problem</u>:

The F-manipulation problem is defined as follows:

Given votes of non-manipulators and a preferred alternative p, can a manipulator cast a vote that makes p uniquely win under F?

Example 3: The computational problem using Borda

If the manipulator is the third voter, the preferred alternative is $p = a$, and the votes of the first two (non-manipulating) voters are given by

1	2
b	b
a	a
c	c
d	d

can voter 3 make a uniquely win under the Borda count? In this case, the answer is yes. Voter 3 can report $a > c > d > b$, resulting in a winning the election with 7 points while b, c and d

lore with 6, 4 and 1 points, respectively.

Consider the following greedy algorithm for the F-manipulation problem:

- rank p in first place
- while there are unranked alternatives:
  - if an alternative can be placed without preventing p from winning, place it
  - otherwise, return FALSE

Consider the same problem as before where voter 3 wants a to win given the other two voters' rankings.
Following the greedy algorithm, we place a first to get:

1	2	3
b	b	a
a	a	
c	c	
d	d	

Now, the algorithm attempts to put b in the second place but making this placement will give b 8 points while a only has 7, so we cannot do that. However, if we place c at the second place, c will only have 4 points, so a still will be the winner. Thus, we have step-by-step ranking process:

1	2	3
b	b	a
a	a	c
c	c	
d	d	

Now, we attempt to place b as the third choice. This would give b 7 points, but since we want a to be the unique winner, we cannot place b here. If we place d in the third place, d will have 1 point and so a will be the winner. Thus, we make this placement and get

1	2	3
b	b	a
a	a	c
c	c	d
d	d	

as our working preference profile. Finally, we attempt to place b in the fourth place of voter 3's ranking, giving b 6 points, which still allows a to win with 7 points. Thus, we make this placement to get the profile

1	2	3
b	b	a
a	a	c
c	c	d
d	d	b

The algorithm concludes after successfully finding a ranking that satisfies the F-manipulation problem.

The Borda count is not the only rule where this greedy procedure works. More generally,

<u>Theorem</u>: Fix $i \in N$ and the votes of other voters. Let F be a rule such that a function $s(R_i, x)$ satisfies:

1) For every R_i, F chooses an alternative that uniquely maximises $s(R_i, x)$

2) If $\{y : y \prec_{R_i} x\} \subseteq \{y : y \prec_{R_i'} x\}$

then $s(R_i, x) \leq s(R_i', x)$

Then, the greedy algorithm decides the F-manipulation problem correctly.

Here, voter i is manipulator. Function s can be thought of as the score function of an alternative under a preference profile. The second criterion just states that if you have two rankings for the manipulator R_i and R_i', and the alternatives ranked below x in R_i are a subset of the alternatives ranked below x in R_i', then the score of x under R_i' is at least as large as the score of x under R_i.

This theorem captures rules like Borda, plurality, Hull's rules, and many other rules that are based on simple scores, as long as you have simple tie-breaking schemes. This is bad news, as we set out to find rules that are computationally hard to manipulate, but a simple greedy algorithm allows these rules to be manipulated.

However, there are a couple of rules that are known to be difficult to manipulate. These include IRV and Hull's rule with more complicated tie-breaking schemes (like second-order Hull scores).
But worst-case hardness is not necessarily an obstacle to manipulation in the average case!